6. The hill in the form of a right triangle has its foot at $(19,3)$. The inclination of the hill to the ground is 45°. Find the equation of the hill joining the foot and top.

7. Find the equation of a line through the given pair of points
   (i) $\left(2,\frac{2}{3}\right)$ and $\left(\frac{-1}{2},-2\right)$ (ii) $(2,3)$ and $(-7,-1)$

8. A cat is located at the point $(-6,-4)$ in xy plane. A bottle of milk is kept at $(5,11)$. The cat wish to consume the milk travelling through shortest possible distance. Find the equation of the path it needs to take its milk.

9. Find the equation of the median and altitude of ΔABC through A where the vertices are $A(6,2)$, $B(-5,-1)$ and $C(1,9)$

10. Find the equation of a straight line which has slope $\frac{-5}{4}$ and passing through the point $(-1,2)$.

11. You are downloading a song. The percent y (in decimal form) of mega bytes remaining to get downloaded in x seconds is given by $y = -0.1x + 1$.
    (i) find the total MB of the song.
    (ii) after how many seconds will 75% of the song gets downloaded?
    (iii) after how many seconds the song will be downloaded completely?

12. Find the equation of a line whose intercepts on the x and y axes are given below.
    (i) $4, -6$ (ii) $-5, \frac{3}{4}$

13. Find the intercepts made by the following lines on the coordinate axes.
    (i) $3x - 2y - 6 = 0$ (ii) $4x + 3y + 12 = 0$

14. Find the equation of a straight line
    (i) passing through $(1,-4)$ and has intercepts which are in the ratio $2:5$
    (ii) passing through $(-8,4)$ and making equal intercepts on the coordinate axes

5.6 General Form of a Straight Line

The linear equation (first degree polynomial in two variables x and y) $ax + by + c = 0$ (where a, b and c are real numbers such that at least one of a, b is non-zero) always represents a straight line. This is the general form of a straight line.

Now, let us find out the equations of a straight line in the following cases

(i) parallel to $ax + by + c = 0$
(ii) perpendicular to $ax + by + c = 0$

5.6.1 Equation of a line parallel to the line $ax + by + c = 0$

The equation of all lines parallel to the line $ax + by + c = 0$ can be put in the form $ax + by + k = 0$ for different values of k.

5.6.2 Equation of a line perpendicular to the line $ax + by + c = 0$

The equation of all lines perpendicular to the line $ax + by + c = 0$ can be written as $bx - ay + k = 0$ for different values of k.

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